Smarter Housing Pricing Through Data

A Statistical and NLP Approach to Analyzing Housing Listing Descriptions

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Why This Project Matters

Housing prices are driven by measurable features

- Size, location, number of rooms explain much of price
- These are the foundation of traditional pricing models

But listing also include descriptive language

- Sellers use terms like “luxury,” “modern,” and “renovated”
- It is unclear whether this language reflects real value or marketing

Opportunity for data analysis

- Listing description contain unstructured text data
- NLP methods can convert text into measurable features
- This allows us to test the impact of language quantitatively

Why it matters in practice

- Helps understand how homes are valued in real markets
- Provides insight into buyer perception vs actual value
- Useful for real estate analytics and housing finance

Business Problem

How can we determine whether listing descriptions influence housing prices beyond structural features?

- Housing markets are complex, and price formation is influenced by demand, which is often unobserved
- Recent research shows that online search behavior can act as a proxy for housing demand and strongly predicts future house prices
- This suggests that non-traditional data sources contain valuable information beyond standard housing features, which may improve price modeling

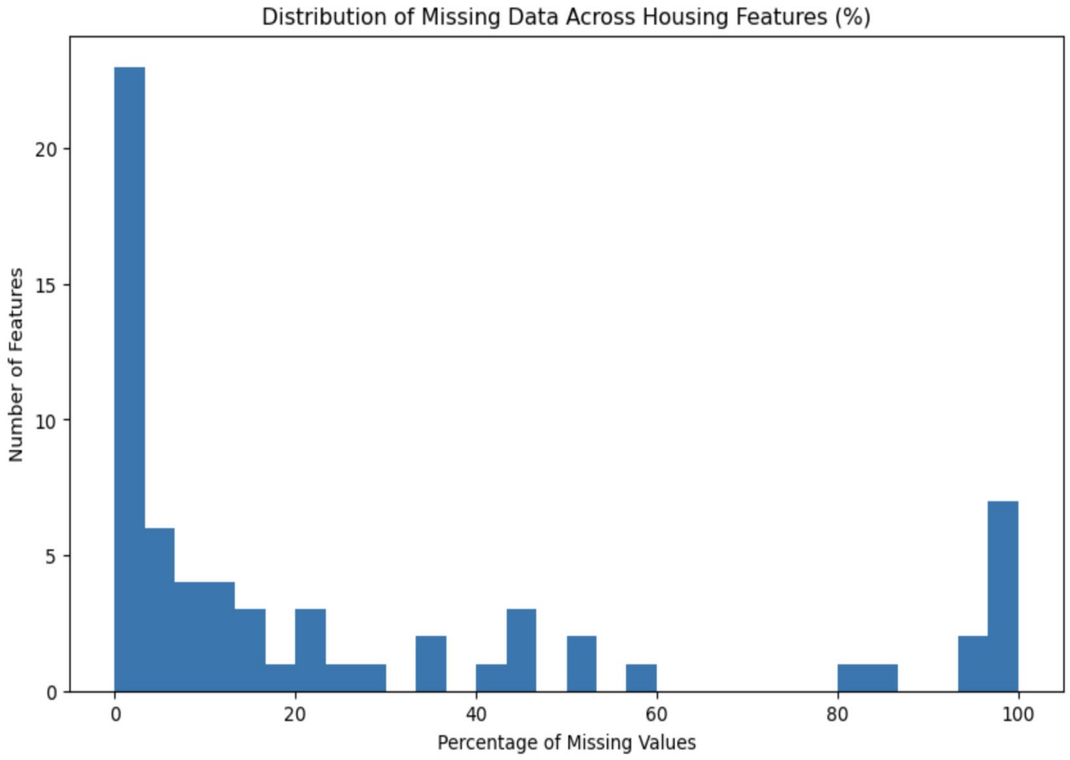
Data Summary

- Name: Virginia Housing Listing Dataset
- Data Source: Wen scraped from Realtor.com using the HomeHarvest Python package
- Scraping Method:
  - Used scrape_property()
  - Filtered for "for_sale" properties in Virginia within the past 30 days
  - Exported results directly to CSV
- Structure: Tabular dataset
- Rows: 9,800
- Features: 66
- Target Variable: log_price (log-transformed listing price)
- Text Feature:
  - Listing description (text)
  - Use for keyword features and TF-IDF modeling

#	Article/Date	Author	Method Used	Improvements to our work
1	“Describe the house and I will tell you the price” (2024)	Zhang et al.	NLP + Regression (TF-IDF, Word2Vec, BERT + ML models)	Shows that text descriptions improve prediction accuracy; our work focuses on statistical inference and significance testing of text features
2	“Using Text Mining to Analyze Real Estate Classifieds” (2015)	Sherief Abdallah	2-stage regression (Structured + TF-IDF text features)	We extend this by focusing on statistical significance (ANOVA + inference) instead of just prediction
3	“Specification and Estimation of Hedonic Housing Price Modelsl” (1991)	Ayse Can	Linear regression (structured features only)	We add textual features to improve traditional hedonic models

Data Cleaning

Handling Missing/Duplicate Data



Dropping Irrelevant Features

- Removed columns with >80% missing values
- Dropped features not related to:
  - Housing structure (price, sqft, baths, etc.)
  - Location
  - Listing Description
- Dropped Duplicates

Handling Missing Data

- Categorical Variables
  - Filled with “Unknown”
- Numerical Variables
  - Filled with median to reduce outlier impact

66 Features → 20 Features

Feature Engineering and Multicollinearity

Feature Engineering

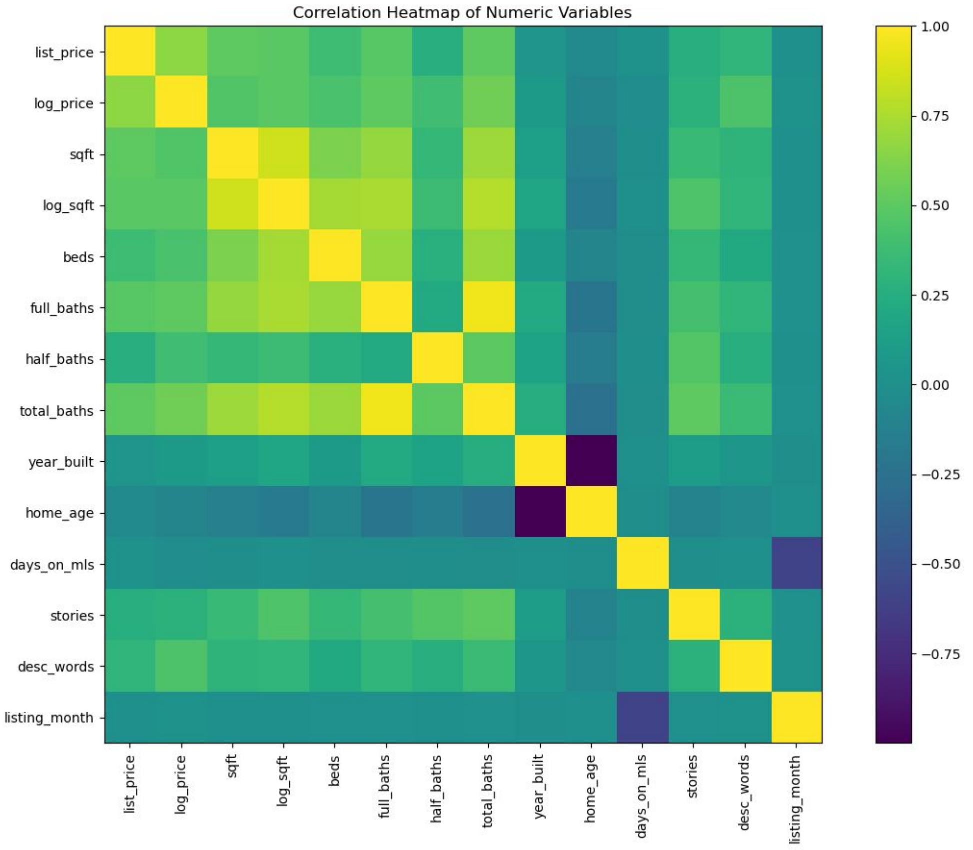
- total _baths = full_baths + half_baths
- home_age = current_year - year_built
- desc_words = number of words in listing description

Text-based features:

- Premium_score
- Layout_score
- Condition_score
- Outdoor_score
- Location_score

Advanced text representation:

- Used TF-IDF to convert descriptions into numerical features



Handling Skew Data

Observed Skewness

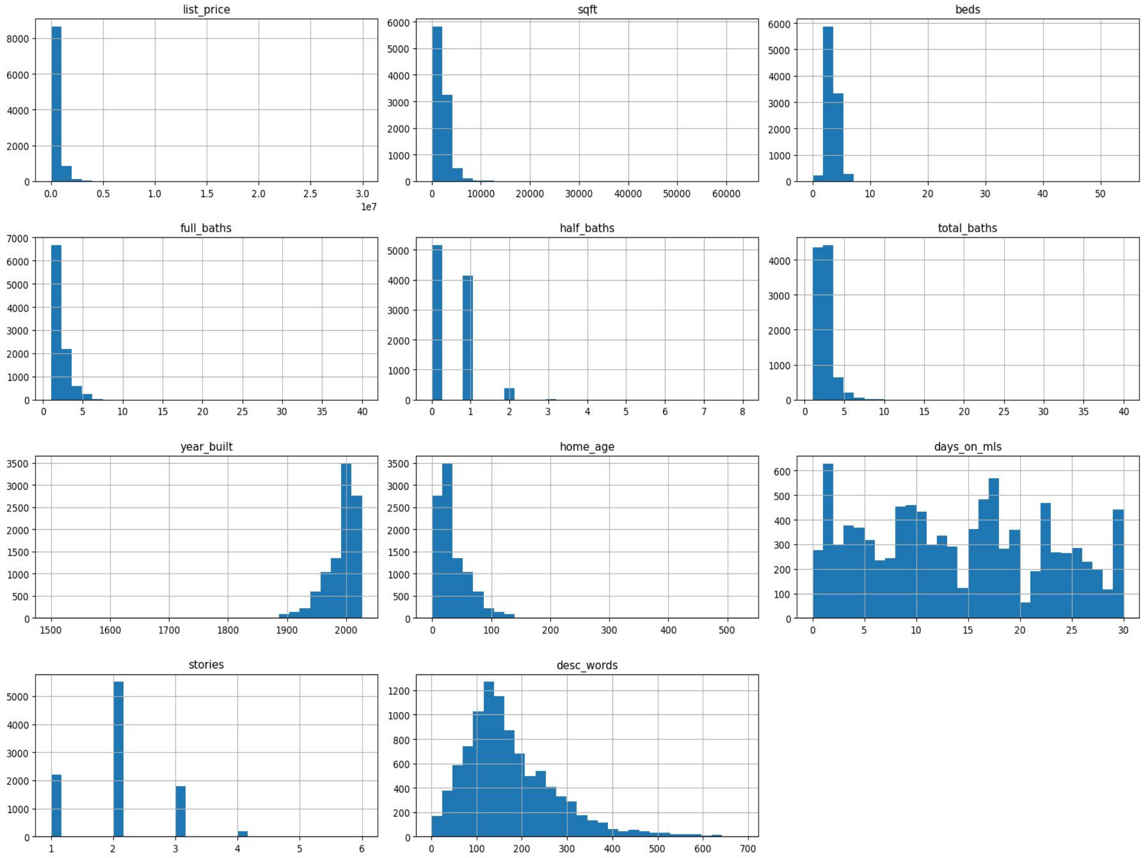
- Many numerical features are right skewed
- Small number of high-value create long tails
- This can violate regression assumption and impact model performance

Transformation Applied

- Applied log transformation

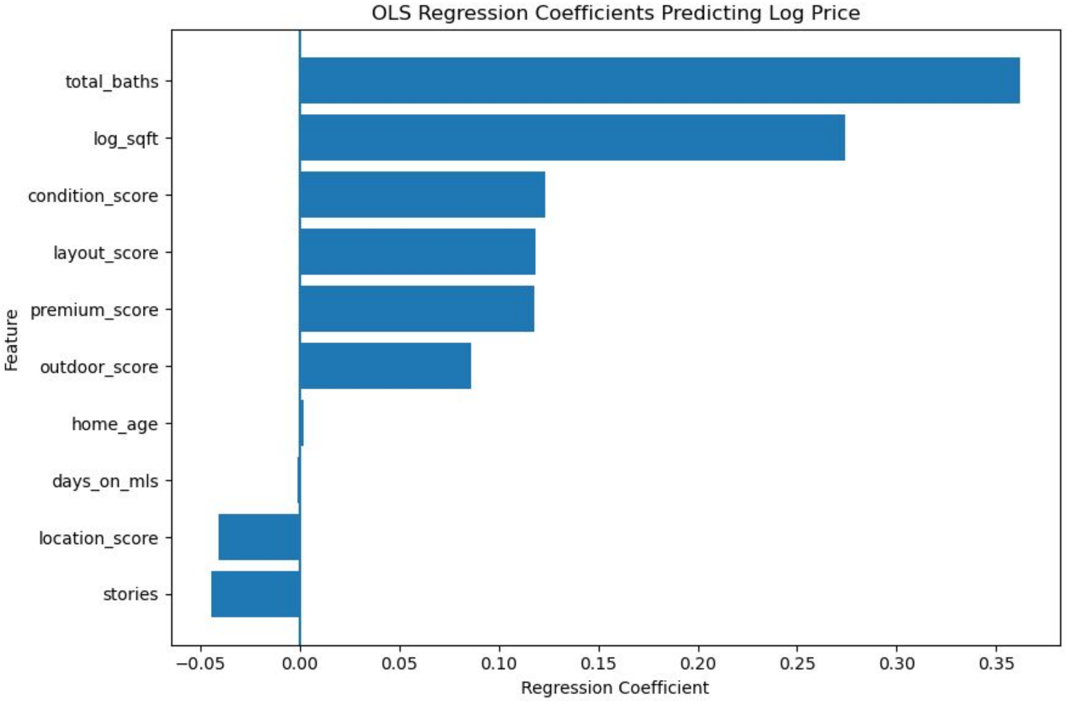
Additional Handling

- Median imputation
- Extreme values retain to preserve real market behavior



Data Modeling

Regression Model – What Drives House Prices



$R^2 = 0.378$

- Total bathrooms (+0.36)
- Square footage (+0.27)
- Text features (premium, layout, condition)

Negative effects:

- Stories (-0.04)
- Location score (-0.04)

Statistical Significance:

- Almost all variables are statistically significant ($p < 0.05$)

Even after controlling for size and structure, **text-based features significantly impact price**

TF-IDF + Ridge (Best Model)

Model Performance

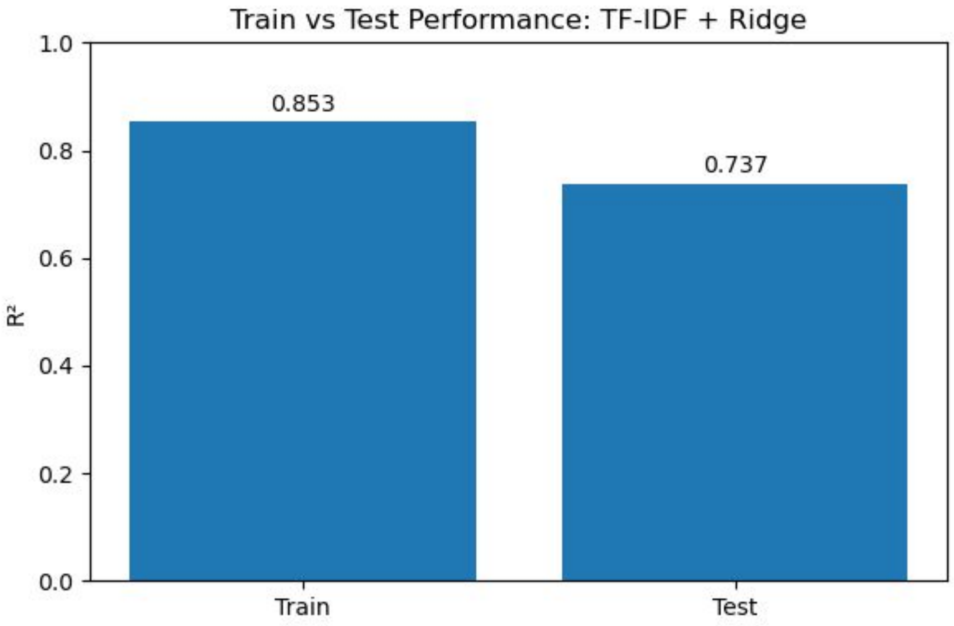
- Train R^2 = 0.85
- Test R^2 = 0.74
- Relatively good generalization

Comparison to Baseline

- Without text: $R^2 \approx 0.57$
- With text: $R^2 \approx 0.74$

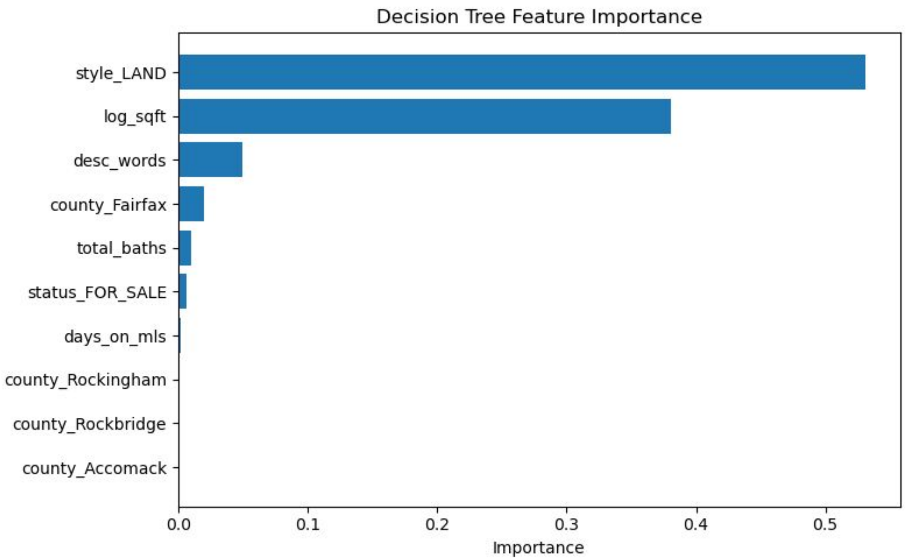
What the Model Does

- Converts text into features using TF-IDF
- Uses Ridge Regression to reduce overfitting
- Combines
  - Structure features (sqft, baths)
  - Text features (listing descriptions)



	Model	R2	RMSE	MAE	R2_Improvement
0	No Text	0.576948	0.608526	0.336181	0.000000
1	With Text	0.737122	0.479688	0.284151	0.160175

Decision Tree



Compared two decision trees:

- Structural features only
- Structural + text features

Text included

- Description length
- Keyword indicators such as luxury, modern, renovated, waterfront

This supports that listing descriptions contain useful pricing information beyond structural housing features.

	Model	R2	RMSE	MAE	R2_Improvement
0	Decision Tree: No Text	0.465753	0.683838	0.415984	0.000000
1	Decision Tree: With Text	0.502577	0.659849	0.403828	0.036824

Text Impact (Key Insight)

Words linked to higher prices:

- Development, acres, estate, waterfront

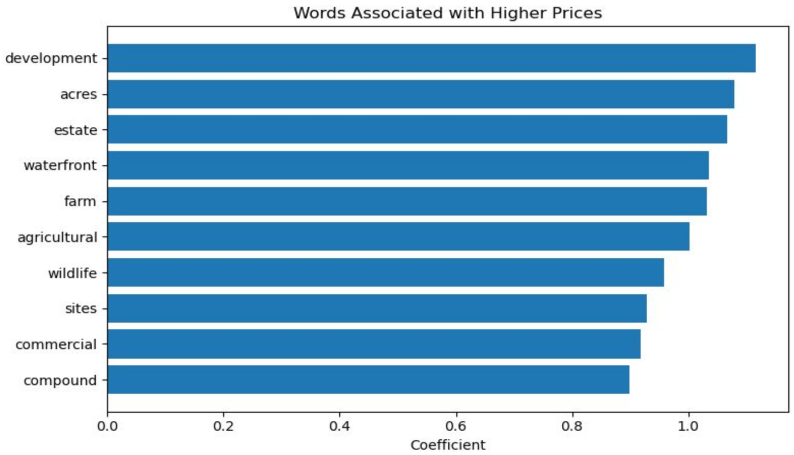
Words linked to lower prices:

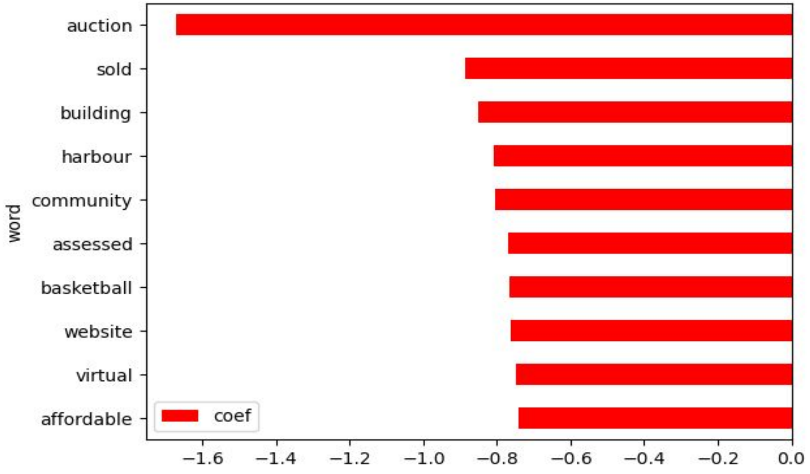
- Auction, sold, building, harbour

“Luxury” keyword test:

- Statistically significant, $p < 0.001$

Listing language contains measurable pricing signals beyond structural features.





Conclusion

How this compares to Literature

- Text only models $R^2 \approx 0.7–0.8$
- Text improves prediction accuracy (lower RMSE)

What Our Models Show

- Baseline model (no text):
    - $R^2 = 0.577$
- Best model (TF-IDF + Ridge)
    - $R^2 = 0.817$

Text features provide substantial predictive power beyond structural features

What This Means

- Even after controlling for:
    - Square footage, bathrooms, home age, etc.
    - **Listing description still significantly impact price**
    - **Specific words matter, text is not just noice, it captures perceived value and marketing signals**

Thank You!